BLEEP Education – Artificial Intelligence Internship
June Batch

STUDENT PERFORMANCE & ACADEMIC ANALYTICS

Exploratory Data Analysis

Presented by: Tripti
Dataset: student-mat.csv
Main Variable: G3 – Final Grade

PROJECT OVERVIEW

- Analyzed 395 student records with 33 features
- 16 numerical + 17 categorical features
- Focused on demographic, academic, family and behavioral attributes
- G3 used as the main indicator of final academic performance

OBJECTIVES

- Understand the structure and quality of the dataset
- Analyze distributions of important student attributes
- Examine associations with final grade (G3)
- Identify numerical relationships using correlation analysis
- Detect potential outliers using the IQR method
- Generate insights through data visualization

Workflow

Data Loading → Data Understanding → Data Quality → EDA → Correlation → Insights

DATA UNDERSTANDING & QUALITY

TABLE 1 : DATASET & DATA QUALITY	
Aspect	Result
Records	395
Features	33
Numerical	16
Categorical	17
Missing Values	0
Duplicate Rows	0
Age Range	15–22
G3 Range	0–20
Absence Range	0–75

Important Variables

- studytime
- failures
- absences
- G1
- G2
- •G3
- Medu
- Fedu
- internet

Outlier Check — IQR

- Age → 1 potential outlier
- Absences → 15 potential outliers
- G1 → 0
- G2 → 13
- G3 → 0

EXPLORATORY DATA ANALYSIS

Univariate Analysis

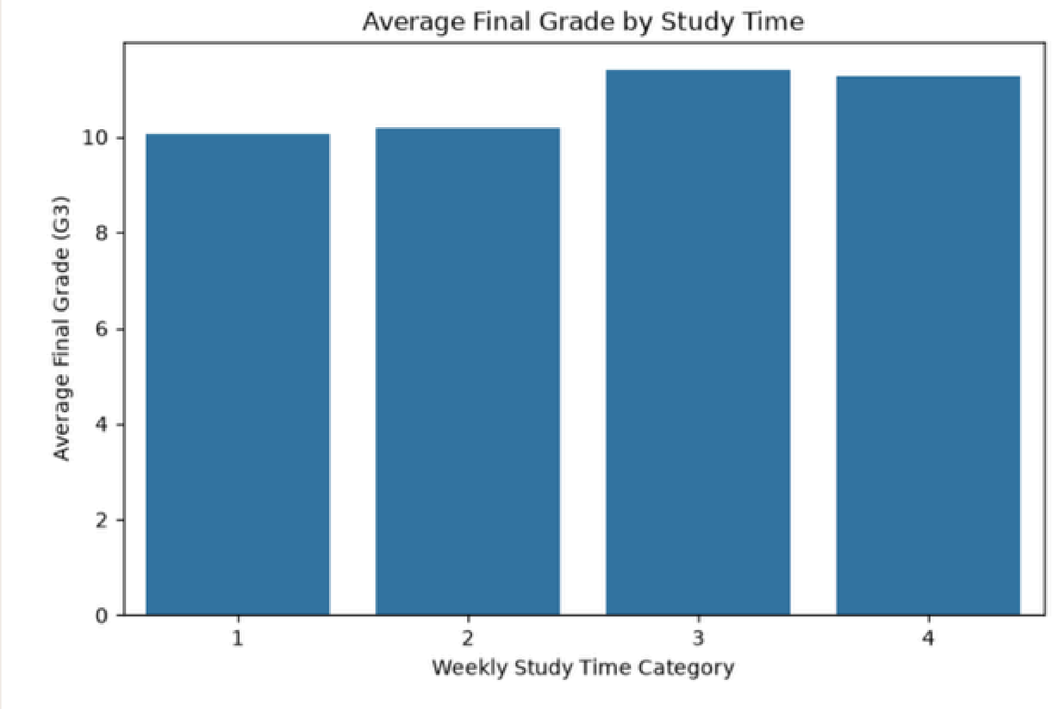
- Final grade distribution
- Absence distribution
- Weekly study-time distribution
- Gender and school distribution

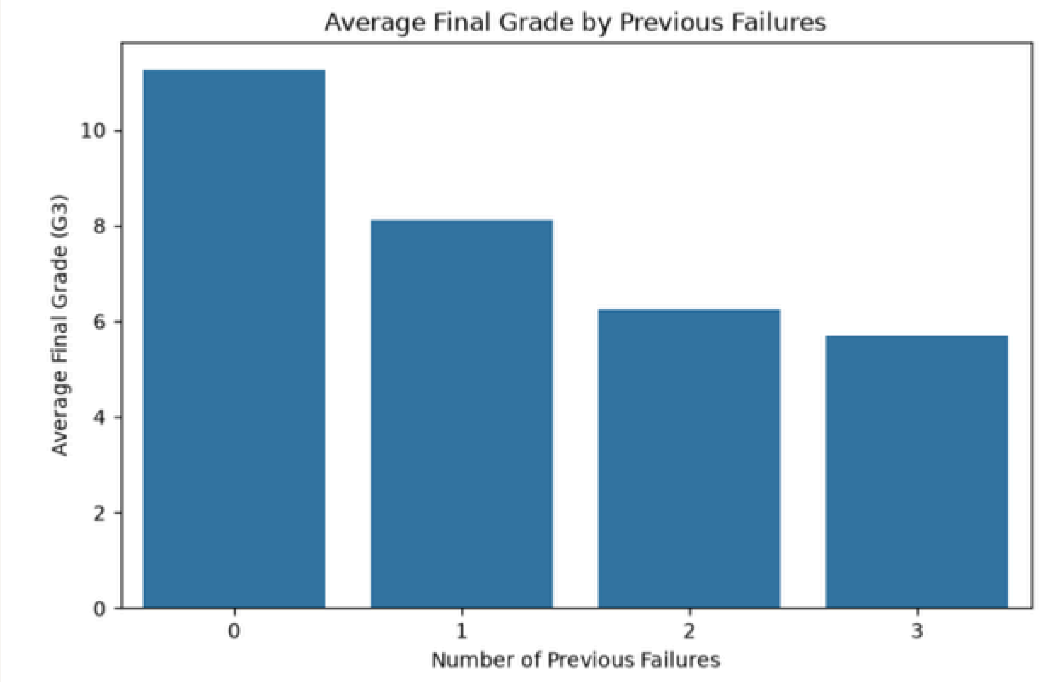
Bivariate Analysis

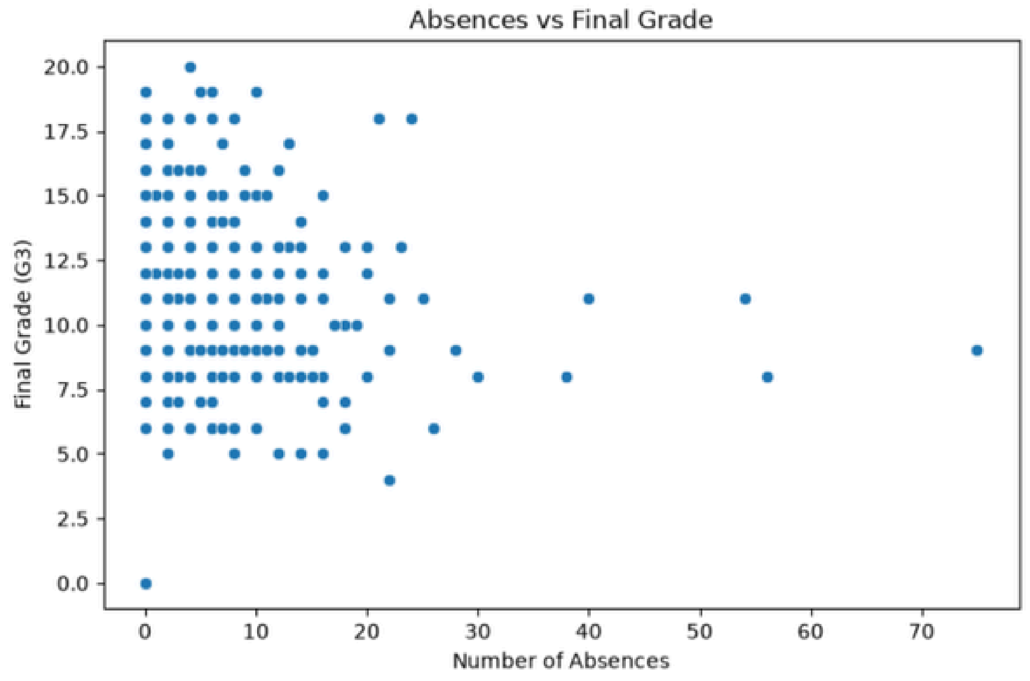
- Gender vs G3
- Study time vs G3
- Previous failures vs G3
- Absences vs G3
- Mother's education vs G3
- Internet access vs G3

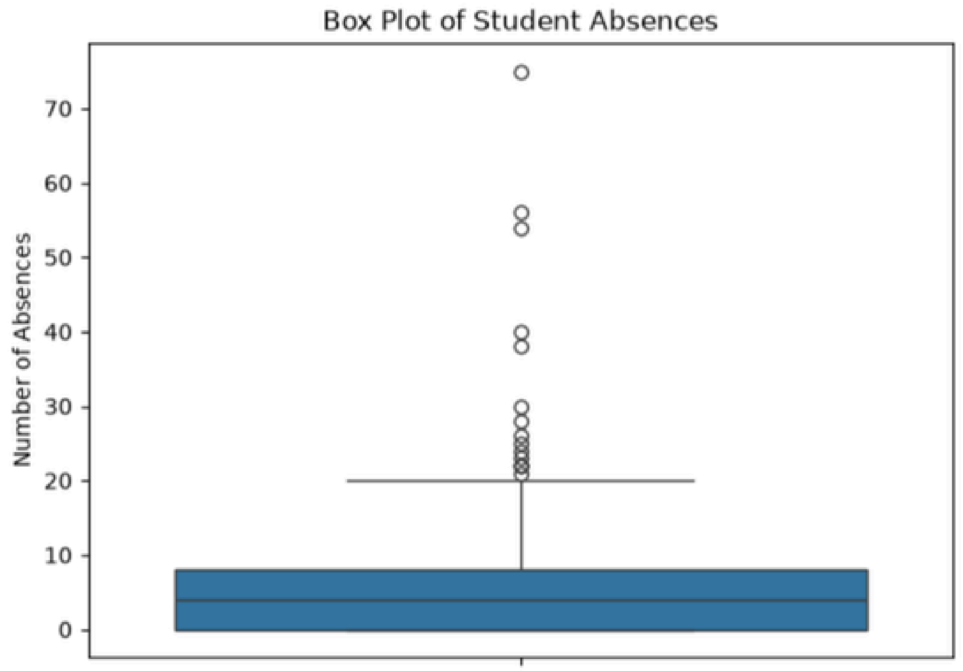
The analysis compares how final grades vary across different demographic, academic and behavioral groups.

EXPLORATORY DATA ANALYSIS


Average Final Grade by Study Time


Average Final Grade by Previous Failures

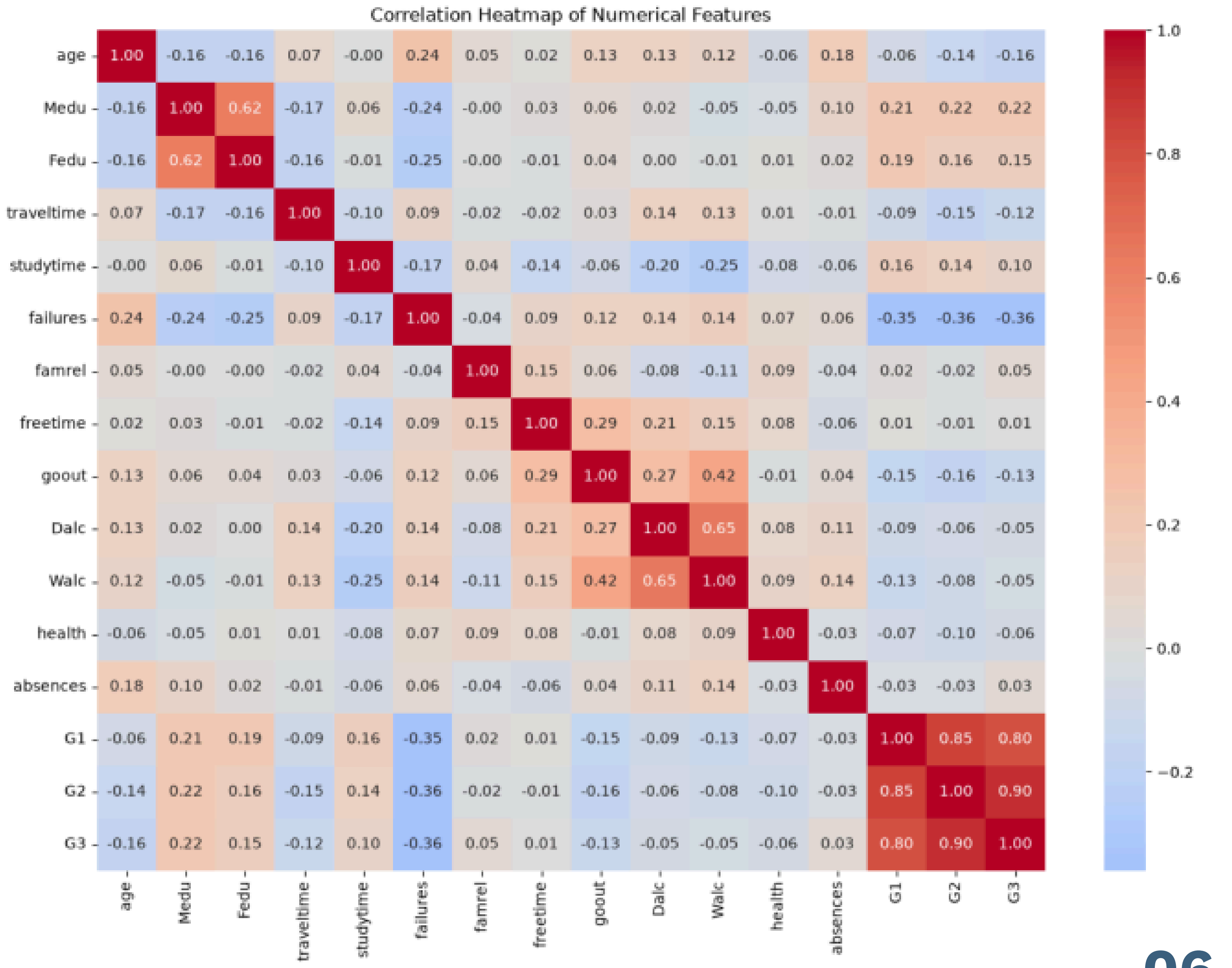

Absences vs Final Grade


Box Plot of Student Absences

These are highlights, not the complete EDA.

CORRELATION ANALYSIS

Variable	Correlation
G2	0.905
G1	0.801
Medu	0.217
Fedu	0.152
Study Time	0.098
Absences	0.034
Failures	−0.360
Age	−0.162



Correlation Heatmap of Numerical Features

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
age	1.00	-0.16	-0.16	0.07	-0.00	0.24	0.05	0.02	0.13	0.13	0.12	-0.06	0.18	-0.06	-0.14	-0.16
Medu	-0.16	1.00	0.62	-0.17	0.06	-0.24	-0.00	0.03	0.06	0.02	-0.05	-0.05	0.10	0.21	0.22	0.22
Fedu	-0.16	0.62	1.00	-0.16	-0.01	-0.25	-0.00	-0.01	0.04	0.00	-0.01	0.01	0.02	0.19	0.16	0.15
traveltime	0.07	-0.17	-0.16	1.00	-0.10	0.09	-0.02	-0.02	0.03	0.14	0.13	0.01	-0.01	-0.09	-0.15	-0.12
studytime	-0.00	0.06	-0.01	-0.10	1.00	-0.17	0.04	-0.14	-0.06	-0.20	-0.25	-0.08	-0.06	0.16	0.14	0.10
failures	0.24	-0.24	-0.25	0.09	-0.17	1.00	-0.04	0.09	0.12	0.14	0.14	0.07	0.06	-0.35	-0.36	-0.36
famrel	0.05	-0.00	-0.00	-0.02	0.04	-0.04	1.00	0.15	0.06	-0.08	-0.11	0.09	-0.04	0.02	-0.02	0.05
freetime	0.02	0.03	-0.01	-0.02	-0.14	0.09	0.15	1.00	0.29	0.21	0.15	0.08	-0.06	0.01	-0.01	0.01
goout	0.13	0.06	0.04	0.03	-0.06	0.12	0.06	0.29	1.00	0.27	0.42	-0.01	0.04	-0.15	-0.16	-0.13
Dalc	0.13	0.02	0.00	0.14	-0.20	0.14	-0.08	0.21	0.27	1.00	0.65	0.08	0.11	-0.09	-0.06	-0.05
Walc	0.12	-0.05	-0.01	0.13	-0.25	0.14	-0.11	0.15	0.42	0.65	1.00	0.09	0.14	-0.13	-0.08	-0.05
health	-0.06	-0.05	0.01	0.01	-0.08	0.07	0.09	0.08	-0.01	0.08	0.09	1.00	-0.03	-0.07	-0.10	-0.06
absences	0.18	0.10	0.02	-0.01	-0.06	0.06	-0.04	-0.06	0.04	0.11	0.14	-0.03	1.00	-0.03	-0.03	0.03
G1	-0.06	0.21	0.19	-0.09	0.16	-0.35	0.02	0.01	-0.15	-0.09	-0.13	-0.07	-0.03	1.00	0.85	0.80
G2	-0.14	0.22	0.16	-0.15	0.14	-0.36	-0.02	-0.01	-0.16	-0.06	-0.08	-0.10	-0.03	0.85	1.00	0.90
G3	-0.16	0.22	0.15	-0.12	0.10	-0.36	0.05	0.01	-0.13	-0.05	-0.05	-0.06	0.03	0.80	0.90	1.00

KEY FINDINGS OF CORRELATION ANALYSIS

- G2 shows the strongest positive linear association with G3.
- G1 also shows a strong positive association with G3.
- Previous failures show a negative association with final grade.
- Mother's and father's education show positive but smaller correlations with G3.
- Absences have a weak linear correlation with G3 in this dataset.
- The heatmap helped compare relationships across all numerical variables.

RESULTS, CONCLUSION & FUTURE SCOPE

Metric	Result
Average G3	10.42
Median G3	11
Highest G3	20
Lowest G3	0
Average Absences	5.71

Conclusion

- EDA provided a clear overview of student performance.
- Identified patterns across study time, failures, absences and background factors.
- Correlation analysis highlighted key numerical associations with G3.
- Provides a foundation for statistical and predictive analysis.

Future Scope

Deeper Statistical Analysis → Feature Engineering → Predictive Modeling → Model Evaluation → Explainable AI

THANK YOU